

Lab02_Pierre

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Exercício 1:

As estatísticas são o número de voos atrasados (`ARRIVAL_DELAY > 10`) e o número total de voos válidos, agrupados por mês, dia e companhia aérea.

Exercício 2:

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

`filter`, `lag`

The following objects are masked from 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

```
getStats <- function(input, pos) {  
  input %>%  
    filter(AIRLINE %in% c("AA", "DL", "UA", "US"),  
           !is.na(ARRIVAL_DELAY)) %>%  
    group_by(MONTH,  
             DAY,  
             AIRLINE) %>%  
    summarise(atrasados = sum(ARRIVAL_DELAY > 10),  
             total = n(), .groups = "drop")  
}
```

Questão 3:

```
library(readr)

stats <- read_csv_chunked(
  "flights.csv",
  DataFrameCallback$new(getStats),
  chunk_size = 100000,
  col_types = cols_only(MONTH = col_integer(),
                        DAY = col_integer(),
                        AIRLINE = col_character(),
                        ARRIVAL_DELAY = col_double())
)
```

Questão 4:

```
computeStats <- function(stats) {
  stats %>%
    group_by(MONTH,
              DAY,
              AIRLINE) %>%
    summarise(atrasados = sum(atrasados),
              total = sum(total),
              .groups = "drop") %>%
    mutate(Cia = AIRLINE,
           Data = as.Date(paste(2015, MONTH, DAY, sep = "-")),
           Perc = atrasados / total) %>%
    select(Cia,
           Data,
           Perc)
}

stats <- computeStats(stats)
```

Questão 5:

```
library(ggplot2)
```

```

pal <- scale_fill_gradient(
  low = "#4575b4",
  high = "#d73027"
)

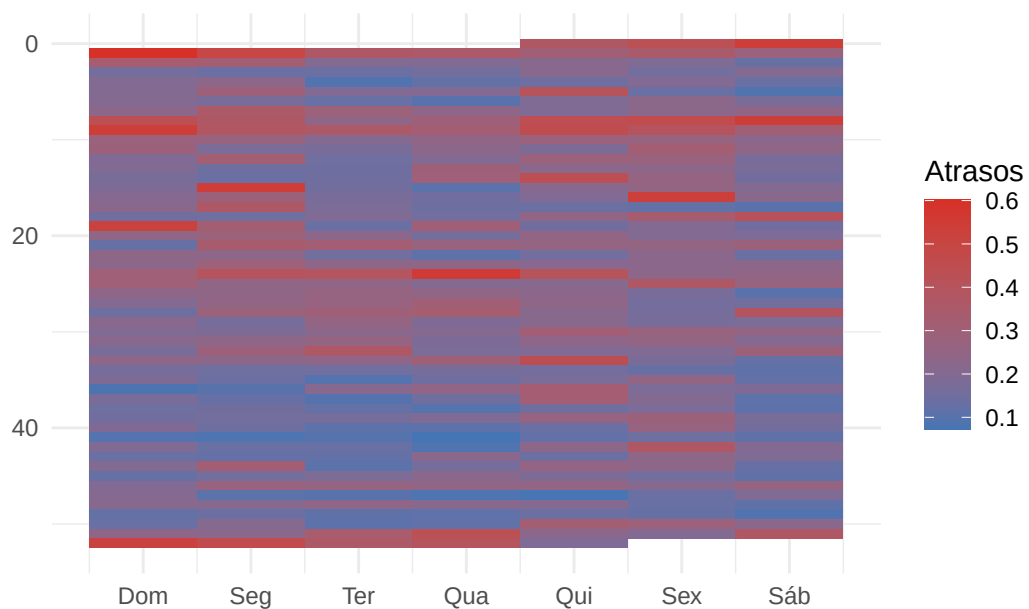
baseCalendario <- function(stats, cia) {
  stats %>%
    filter(Cia == cia) %>%
    mutate(semana = as.numeric(format(Data, "%U")),
           dia_semana = as.numeric(format(Data, "%w")))
}

cAA <- baseCalendario(stats, "AA")
cDL <- baseCalendario(stats, "DL")
cUA <- baseCalendario(stats, "UA")
cUS <- baseCalendario(stats, "US")

ggplot(cAA,
       aes(x = dia_semana,
           y = semana,
           fill = Perc)) +
  geom_tile() +
  pal +
  scale_x_continuous(breaks = 0:6,
                    labels = c("Dom", "Seg", "Ter", "Qua", "Qui", "Sex", "Sáb")) +
  scale_y_reverse() +
  ggtitle("Percentual de voos atrasados - AA") +
  labs(x = NULL,
       y = NULL,
       fill = "Atrasos") +
  theme_minimal()

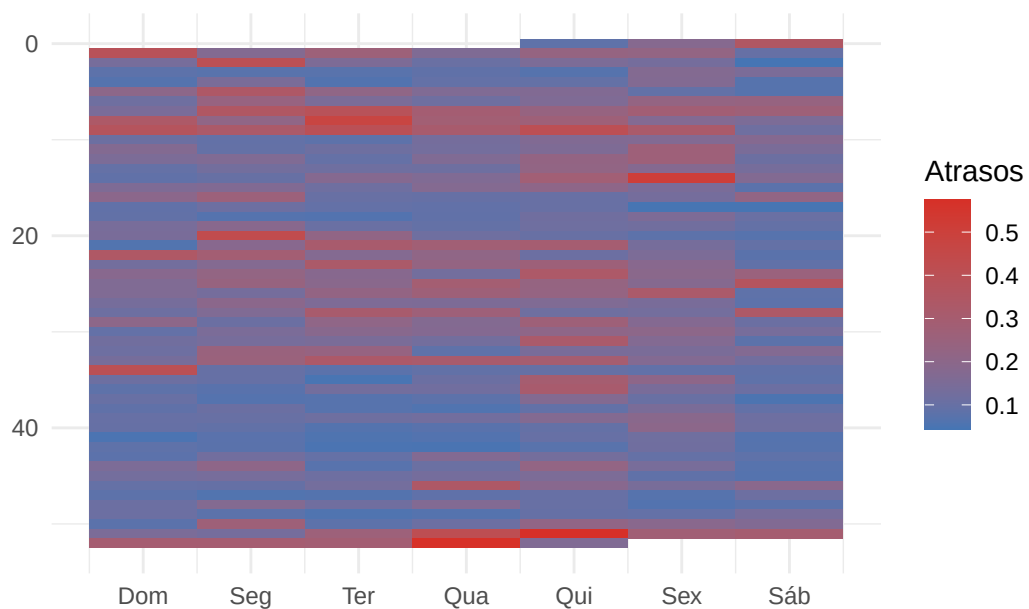
```

Percentual de voos atrasados – AA



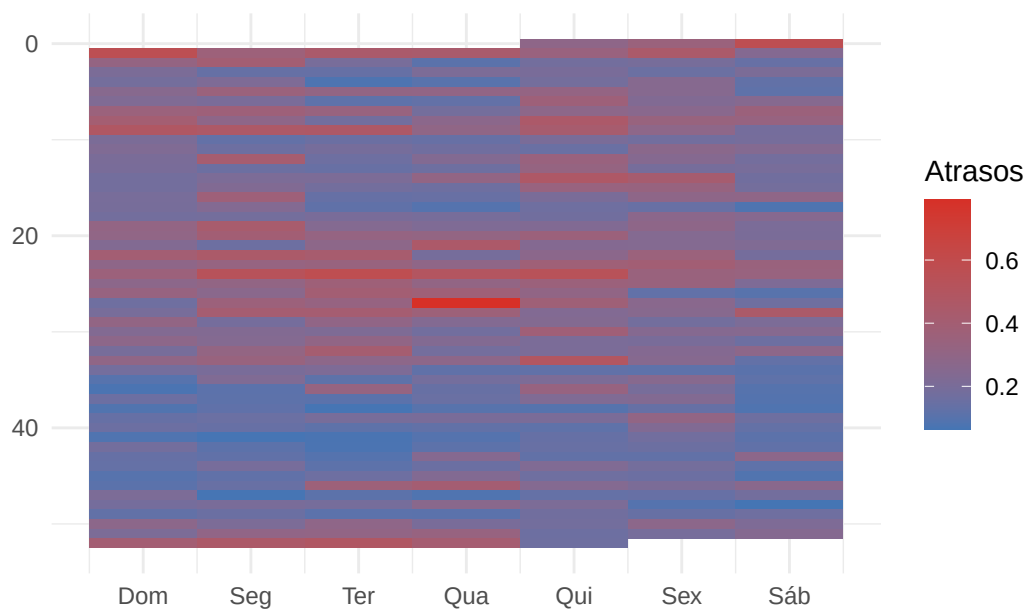
```
ggplot(cDL, aes(x = dia_semana,
                y = semana,
                fill = Perc)) +
  geom_tile() +
  pal +
  scale_x_continuous(breaks = 0:6,
                    labels = c("Dom", "Seg", "Ter", "Qua", "Qui", "Sex", "Sáb")) +
  scale_y_reverse() +
  ggtitle("Percentual de voos atrasados – DL") +
  labs(x = NULL,
       y = NULL,
       fill = "Atrasos") +
  theme_minimal()
```

Percentual de voos atrasados – DL



```
ggplot(cUA, aes(x = dia_semana,
                y = semana,
                fill = Perc)) +
  geom_tile() +
  pal +
  scale_x_continuous(breaks = 0:6,
                    labels = c("Dom", "Seg", "Ter", "Qua", "Qui", "Sex", "Sáb")) +
  scale_y_reverse() +
  ggtitle("Percentual de voos atrasados - UA") +
  labs(x = NULL,
       y = NULL,
       fill = "Atrasos") +
  theme_minimal()
```

Percentual de voos atrasados – UA



```
ggplot(cUS, aes(x = dia_semana,
                 y = semana,
                 fill = Perc)) +
  geom_tile() +
  pal +
  scale_x_continuous(breaks = 0:6,
                     labels = c("Dom", "Seg", "Ter", "Qua", "Qui", "Sex", "Sáb")) +
  scale_y_reverse() +
  ggtitle("Percentual de voos atrasados - US") +
  labs(x = NULL,
       y = NULL,
       fill = "Atrasos") +
  theme_minimal()
```

Percentual de voos atrasados – US

